KO=0.5 Native-VC Brill Campaign and Experimental CPBC Assessment

AthenaK investigation handoff

26 August 2026

Verdict

KO=0.5 does not yield a stable, convergent Figure-3 collapse evolution. The matched-tree N128/N256/N512 campaign is near fourth order early, then loses convergence first outside $r \leq 4M$ around the $\rho \simeq 5M$ mode. N256 fails at $t = 14.405106M$ with an indefinite conformal metric after a catastrophic noncentral runaway. N512 reaches the forced common stop but is also severely unstable. The nonmonotone resolution ordering rules out convergence.

The common numerical curves end at central proper time 8.827, before the published first peak near 10.31 and deep minimum near 12.62. Figure 3 is not reproduced. The Cartoon constraint measure already uses $2\pi\rho\sqrt{\gamma}\,d\rho\,dz$, so the jumps are not a collapsed-y normalization artifact.

Matched-tree outcomes

Case	C^2 inventory	max $\lvert\mathcal{K}\rvert$	Δt
N128	2.344×10^{1}	9.441×10^{1}	2.888×10^{-8}
N256	2.677×10^{18}	8.434×10^{30}	5.615×10^{-9}
N512	1.774×10^{5}	4.244×10^{9}	3.081×10^{-9}

All 141 noninitial hierarchy events replayed exactly. Early regional constraint orders are close to four. At $t = 8M$, $r \leq 4M$ remains near fourth order while $r \leq 8M$ has already degraded, localizing the first loss of convergence outside the center.

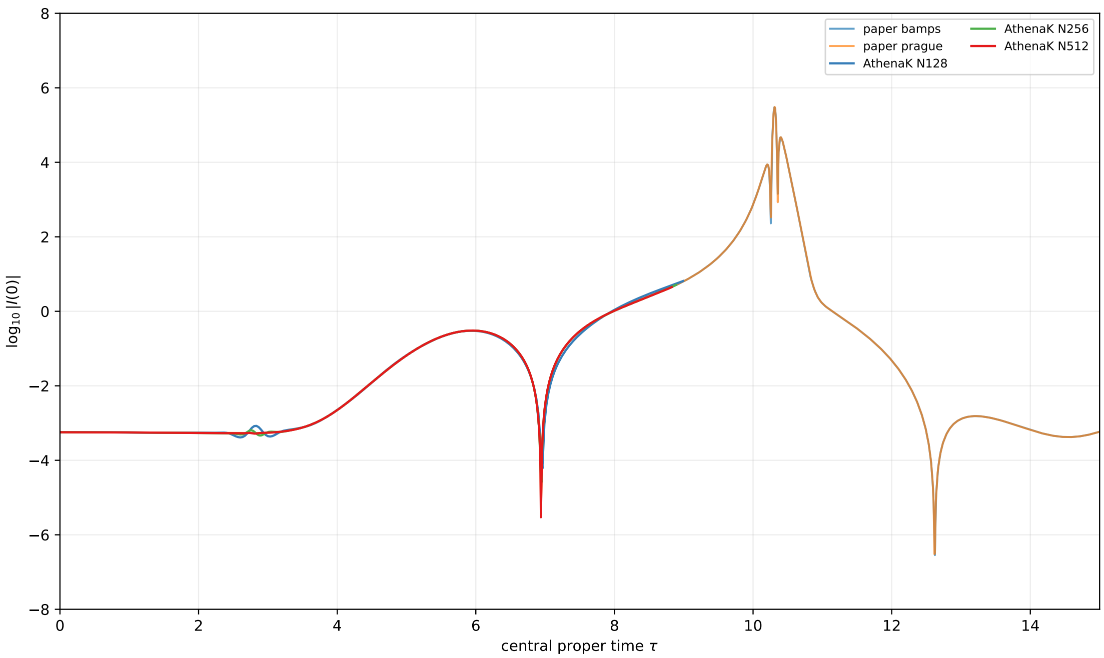

Figure 1: Published Figure-3 curves and the three AthenaK resolutions. No shifts or rescalings are applied. The AthenaK interval ends before the first peak.

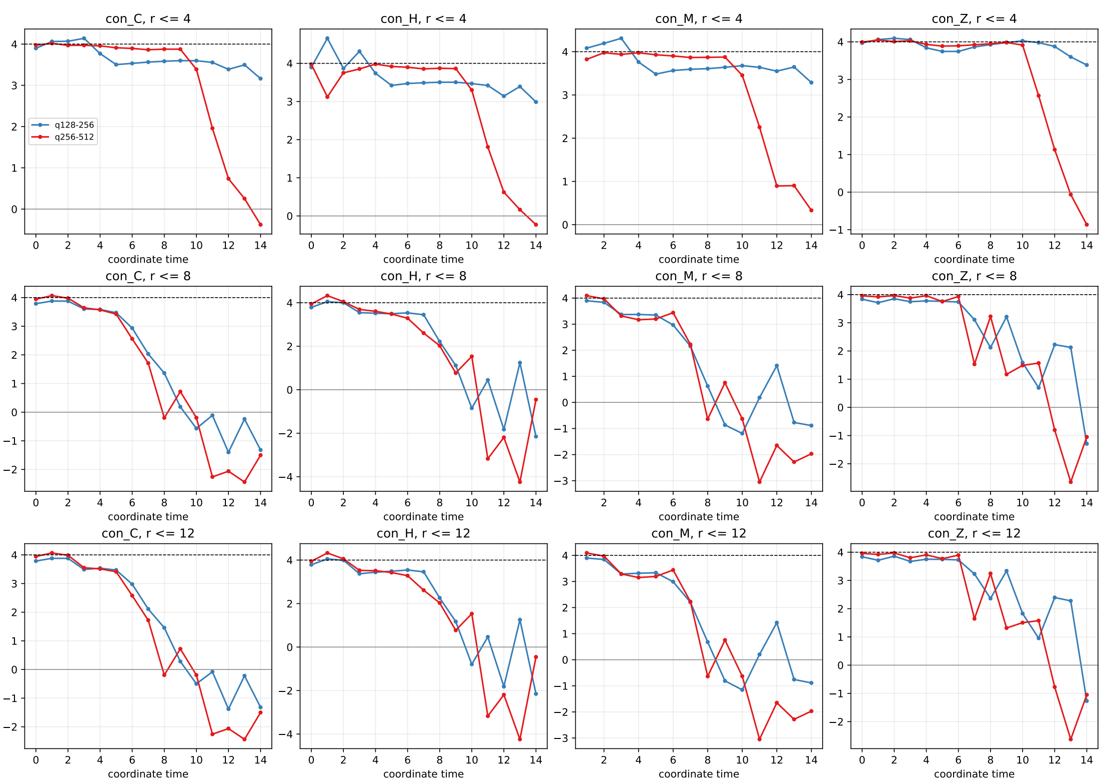

Figure 2: Pairwise amplitude orders on the matched common lattice. The diagnostic filters scales below $h = 0.25$ and is not the native leaf quadrature.

Experimental Bjorhus compatibility boundary

A concrete frame-construction bug was found and repaired: raised normal components consumed not-yet-initialized covariant components for non-diagonal metrics. Commit `d39822c6522688749fe5ead8025907bc055f02f8` separates initialization and index raising and adds a regression test. The strict residual gate was not weakened, and final CUDA tests pass.

The planned A/B/C/D discriminator is incomplete. A (Rout16 original) and C (Rout128 original) reached $t = 6.5M$. B (Rout16 CPBC) developed a corner-local runaway and failed at $t = 3.244461M$. D was intentionally cancelled at $t = 1.508791M$ and is excluded. The current CPBC is not production-stable, and no claim of boundary-error reduction is supported.

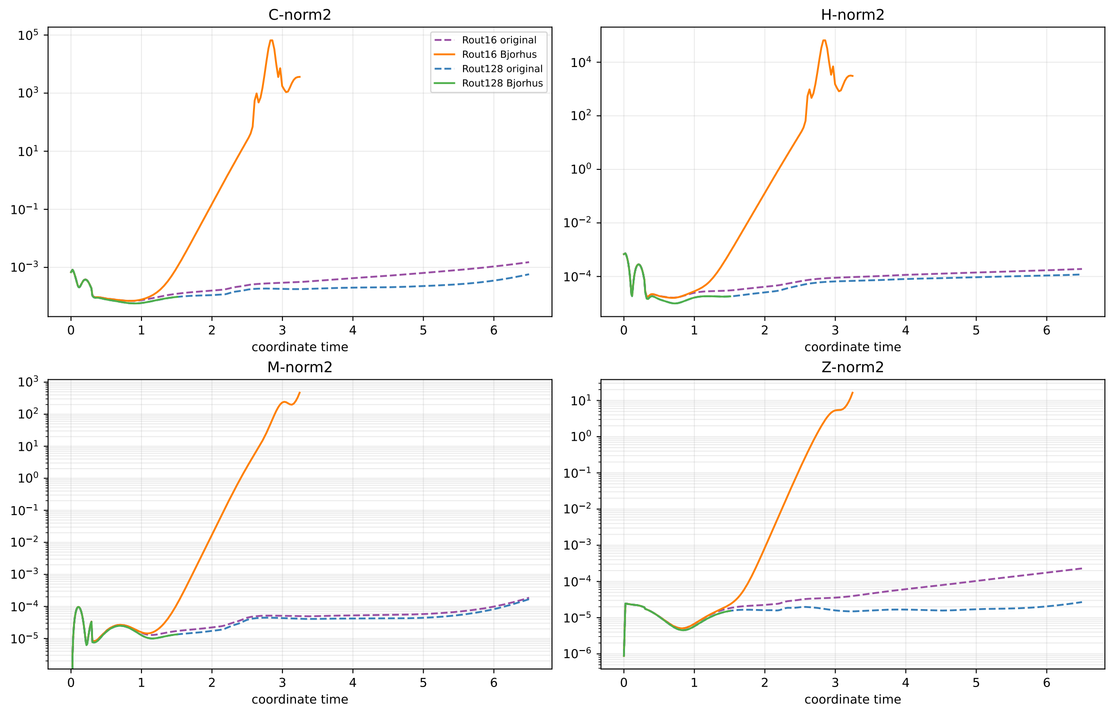

Figure 3: Retained boundary discriminator histories. D is partial and excluded; the four-way comparison is not qualified.

Claim boundary and next step

Current evidence strongly deprioritizes the physical outer boundary and Cartoon axis relative to a bulk/AMR high-frequency or interface mechanism near $\rho \simeq 5M$, but does not mathematically exclude boundary effects or isolate a unique source bug. No t=30 stability, Figure-3 reproduction, convergence, horizon, or critical-phenomena claim is made. The smallest next action should be a bounded stage-resolved diagnostic around the first loss of regional convergence, not a long evolution or broad parameter sweep.

Full paths, hashes, limitations, and compact CSV evidence are recorded in `EVIDENCE_MANIFEST.json` and the accompanying Markdown reports.